

Day 02 - Docker Images & Dockerfile

Objective

Understand Docker Images, Dockerfiles, image layers, UnionFS, Docker build cache, and production workflows.

Topics Covered

Docker Images, Image vs Container, Dockerfile, Build Process, Layers, UnionFS, Build Cache, FROM, WORKDIR, COPY, ADD, RUN, CMD, ENTRYPOINT, ENV, EXPOSE, .dockerignore, docker build, docker image inspect, docker history.

Image Workflow

Dockerfile -> Build -> Docker Image -> Run -> Container.

Commands

docker build -t my-app .; docker images; docker image inspect my-app; docker history my-app; docker pull nginx; docker rmi my-app.

Best Practices

Use official images, pin versions, optimize layer caching, use .dockerignore, avoid latest tag, never store secrets inside images.

Labs

Build your first Docker image, inspect it, view image history, run a container, modify HTML, rebuild and observe cache.

Interview Questions

What is a Docker Image? Explain image layers. Difference between COPY and ADD. Difference between CMD and ENTRYPOINT. What is Docker build cache?